1.I1 – Molecular Structure

Answer parts a) **AND** b) and **EITHER** part c) or d) of this question.

a) Answer **ALL** parts of this question.

i) What are the ions present in the compound $[NH_4][PCl_3F_3]$?

(1 mark)

ii) Using VSEPR theory, predict the pseudo-structure and structure of each ion (include a drawing of both structures in your answer).

(4 marks)

iii) Does either of the two ions possess stereoisomers? If so, draw the structures of these isomers.

(1 mark)

b) Answer **ALL** parts of this question.

i) Sketch and label the molecular orbital (MO) energy level diagram for NO. On your diagram, include drawings of the MOs and electron occupancies. Indicate if significant MO mixing is expected, highlighting which molecular orbitals are likely to be involved.

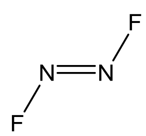
(10 marks)

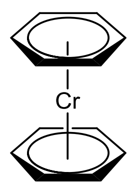
ii) Using your diagram rationalise the following data: the dissociation energy for NO is 627 kJ mol^{-1} whilst that of NO^+ is 1048 kJ mol^{-1}.

(3 marks)

c) Answer **ALL** parts of this question.

i) Indicate the principal axis of symmetry and determine the point group for each of the following molecules:

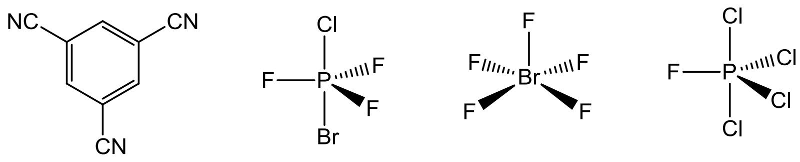




(4 marks)

QUESTION CONTINUED OVERLEAF

ii) For the following molecules, indicate if they possess a horizontal plane of symmetry (σ_h):



(2 marks)

d) Answer **ALL** parts of this question:

i) Using the VSEPR theory, predict whether HCN is linear or bent.

(2 marks)

ii) What is the hybridization of the C atom?

(1 mark)

iii) Applying Valence Bond theory, show a bonding scheme using the hybridization you have suggested (including sketches of the orbitals involved).

(3 marks)